AI/LLM QUALITY ENGINEERING

ENTERPRISE EVALUATION FRAMEWORK

”Looking Correct” Isn’t Enough: How to Qualify AI & LLM Responses

If your organization is adopting AI, this framework ensures outputs aren't just plausible—but complete, grounded, consistent, and production-ready.

CORE CAPABILITY
6-Dimension Response Qualification

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GOVERNANCE
Mandatory Release Gates

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EXECUTIVE SUMMARY

The Core Discovery: The Completeness Trap

Correctness Is Not Quality

Factually true responses frequently create operational risks by omitting mandatory conditions.

Multi-Dimensional Assessment

Evaluates accuracy, groundedness, relevance, instruction following, completeness, and consistency.

Engineering Governance

Weighted averages are combined with mandatory pass thresholds to prevent hidden failures.

⚠ KEY CASE STUDY FINDING

"Models consistently returned 100% correct facts while omitting mandatory approval steps."

A response can pass standard factual accuracy checks and still fail production readiness due to incomplete instruction execution.

PARADIGM SHIFT

Deterministic Testing vs. LLM Quality Engineering

TRADITIONAL QA (DETERMINISTIC)

Input → Expected Output → Pass/Fail

Binary evaluation based on exact matching and deterministic system behavior.

LLM QA (NON-DETERMINISTIC)

Prompt → Multi-Dimensional Evaluation

Continuous scoring across varied phrasing, completeness, and reasoning stability.



Omitted Details

Factually true but misses critical constraints.



Unsupported Claims

Answers questions using unverified assumptions.



Format Violations

Follows topic intent but breaks output structure.



Run Instability

Produces shifting conclusions across identical runs.

MEASUREMENT MATRIX

The 6 Evaluation Dimensions

Accuracy
25% WEIGHT

Factual correctness of statements and conclusions.

Groundedness
20% WEIGHT

Claims strictly supported by provided source data.

Relevance
15% WEIGHT

Direct response without unnecessary drift or fluff.

Instruction Following
15% WEIGHT

Adherence to formatting, constraints, and structure.

Completeness
15% WEIGHT

Inclusion of all required details and criteria.

Consistency
10% WEIGHT

Stability of answer meaning across 5 repeated runs.

SCORING STANDARDS

Standardized 1–5 Scoring Rubric

SCORE	LEVEL	QUALITATIVE EVALUATION CRITERIA
5	**Exemplary**	Completely accurate, fully grounded, perfectly structured, with zero missing conditions.
4	**Minor Variance**	Mostly accurate and complete. Minor non-critical formatting or detail omission.
3	**Partial Defect**	Noticeable accuracy uncertainty, missing key requirements, or moderate instruction drift.
2	**Substantial Failure**	Multiple factual errors, ungrounded assumptions, or major instruction violations.
1	**Critical Failure**	Severe hallucination, direct factual contradiction, or total failure to execute instructions.

RELEASE GOVERNANCE

Mandatory Quality Gate Architecture

PASS FORMULA

PASS = Weighted Score ≥ 3.5 AND (Accuracy ≥ 4 & Groundedness ≥ 4 & Instructions ≥ 4 & Completeness ≥ 4)

Weighted Threshold

Overall score must reach at least 3.50 out of 5.00 across all weighted dimensions.

Mandatory Dimension Floors

Accuracy, Groundedness, Instructions, and Completeness must individually score ≥ 4.

QA PROCESS

8-Step Test Execution Workflow

STEP 01
Define Objective
Target specific risk vector.

STEP 02
Define Baseline
Set ground truth response.

STEP 03
Execute 5× Runs
Test prompt repeatedly.

STEP 04
Capture Output
Record unedited responses.

STEP 05
Apply Rubric
Score 6 quality dimensions.

STEP 06
Compute Score
Calculate weighted total.

STEP 07
Apply Gate
Enforce PASS/FAIL criteria.

STEP 08
Log Evidence
Document score rationale.

EMPIRICAL FINDINGS

Test Coverage Results Summary

TEST ID	QUALITY FOCUS AREA	EXECUTION STATUS	WEIGHTED SCORE	RESULT
TC01	Factual Accuracy Verification	Actual Execution	**5.00 / 5.00**	PASS
TC02	Information Gap / Hallucination Handling	Actual Execution	**5.00 / 5.00**	PASS
TC03	Instruction Format Adherence	Actual Execution	**5.00 / 5.00**	PASS
TC04	**Completeness Evaluation**	Actual Execution	**4.15 / 5.00**	FAIL
TC05	Groundedness Violation	Illustrative Model	**2.30 / 5.00**	FAIL
TC06	Consistency & Stability Failure	Illustrative Model	**4.10 / 5.00**	FAIL

CASE STUDY DEEP DIVE

TC04 Defect Analysis: Completeness

Policy Constraints (Input)

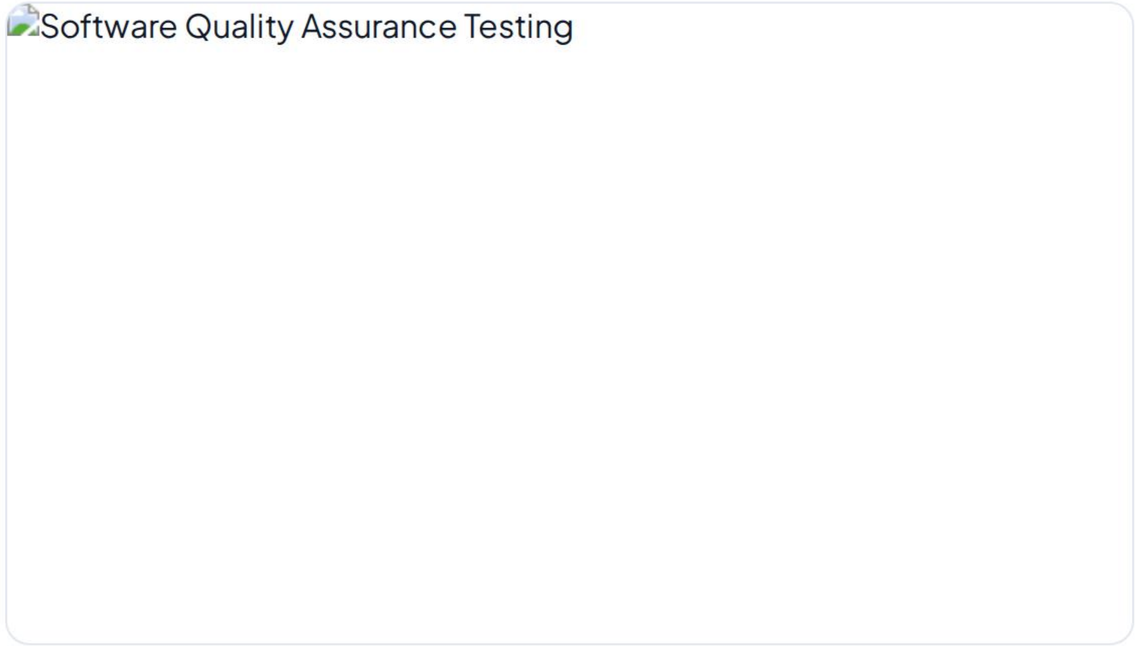
1. Close Critical Defects. 2. Complete Regression. 3. Obtain Manager Approval.

Observed LLM Output

Returned conditions #1 and #2 correctly, but repeatedly omitted condition #3.

Quality Gate Outcome: REJECTED

Weighted Score = 4.15, but Completeness = 3. Fails mandatory floor requirement (Completeness ≥ 4).



FAILURE SCENARIOS

Illustrative Failure Analysis

TC05 — GROUNDEDNESS

SCORE: 2.30

Hallucinated Responsibility

Source policy omitted team assignment. Model claimed: "QA team performs testing."

QA Takeaway:
Plausible answers are unacceptable when unbacked by source documents.

TC06 — CONSISTENCY

SCORE: 4.10

Contradictory Conclusions

5 repeated runs yielded alternating approvals and denials for deployment.

QA Takeaway:
High weighted score (4.10) masked low consistency (1/5). Gate rejected release.

DELIVERABLES

Reusable Enterprise QA Artifacts

Test Scenario Catalog
Structured prompts, risk vectors, and targets.

LLM Evaluation Matrix
Per-run scores, evaluator logs, and evidence.

Evaluation Rubric
Standardized criteria across 6 dimensions.

Quality Gate Rules
Automated release pass/fail threshold logic.

Eval AI – Empowering the Future of Monitoring and Evaluation

Eval AI is a cutting-edge platform designed to revolutionize the monitoring and evaluation (M&E) field by harnessing the power of artificial intelligence. As part of EvalCommunity, Eval AI leverages advanced algorithms and machine learning to streamline data analysis, automate reporting, and provide real-time insights, enabling M&E professionals to make more informed decisions.



QA LEADERSHIP VISION

Next-Phase AI Quality Engineering Roadmap

PHASE 1 (COMPLETE)

Manual POC Baseline

Established 6-dimension rubric, human scoring matrix, and initial quality gates.

PHASE 2 (NEAR-TERM)

Expanded Risk Vectors

Testing RAG groundedness, prompt injection, and context-window boundaries.

PHASE 3 (SCALE)

Automated CI/CD Gates

Integrating LLM-as-a-Judge pipelines directly into deployment workflows.

"Goal: Move from subjective impressions ('looks good') to empirical, repeatable AI quality evidence."

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Image Sources



https://www.inflectra.com/GraphicsViewer.aspx?url=~/Ideas/Topics/Software-Quality-Assurance.doc&name=wordml://03000004.png

Source: www.inflectra.com



https://www.evalcommunity.com/wp-content/uploads/2025/05/Eval-AI.png

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